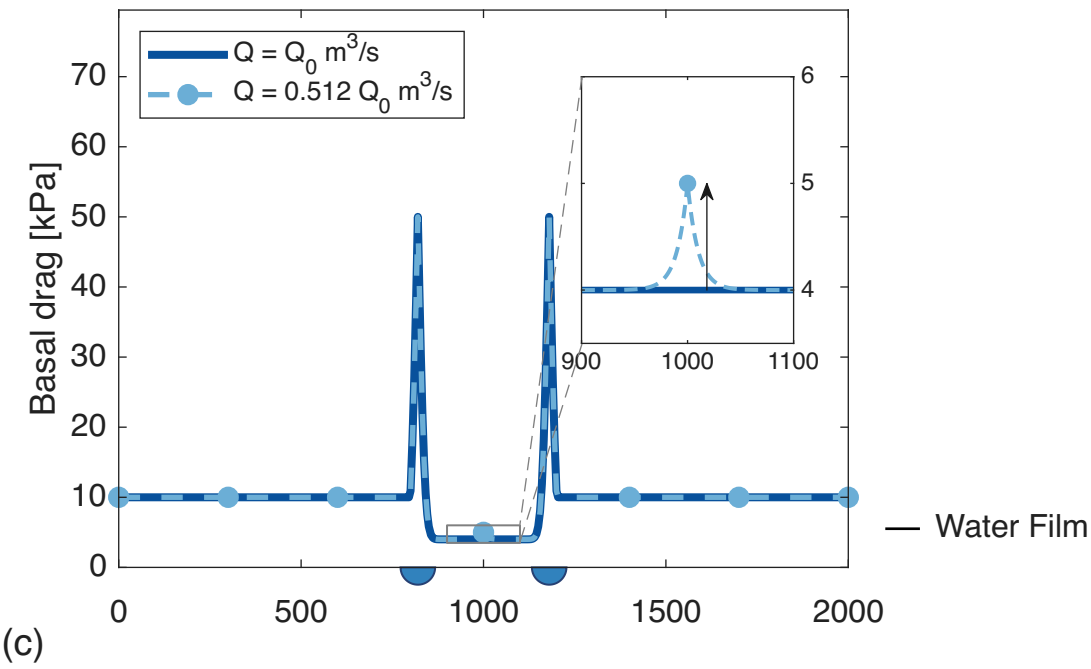
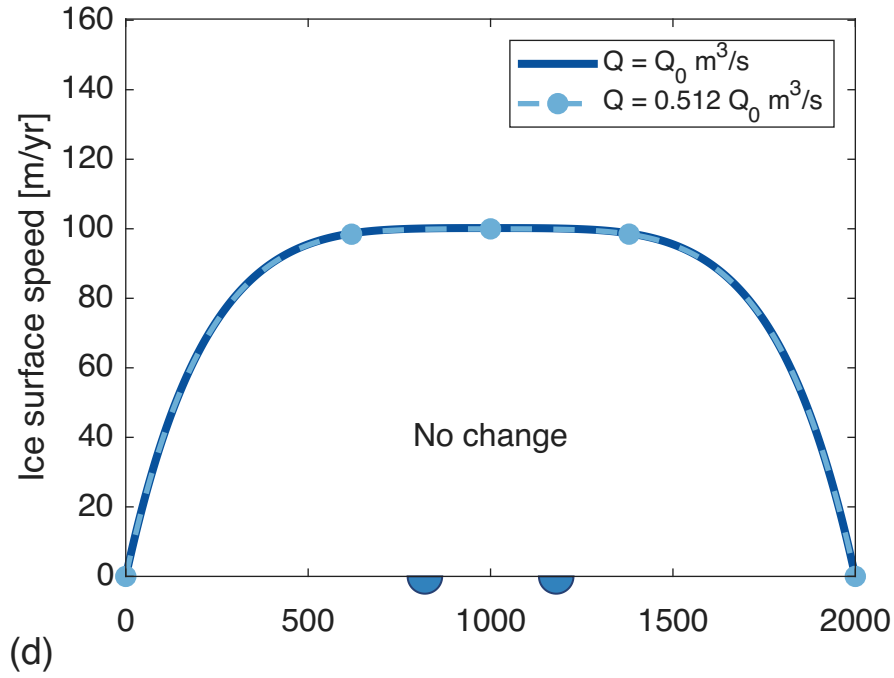
(a)


Q = Q0 m3/s
Q = 0.512 Q0 m3/s
Basal drag [kPa]
0
10
20
30
40
50
60
70
0
500
1000
1500
2000
900
1000
1100
4
5
6
Water Film

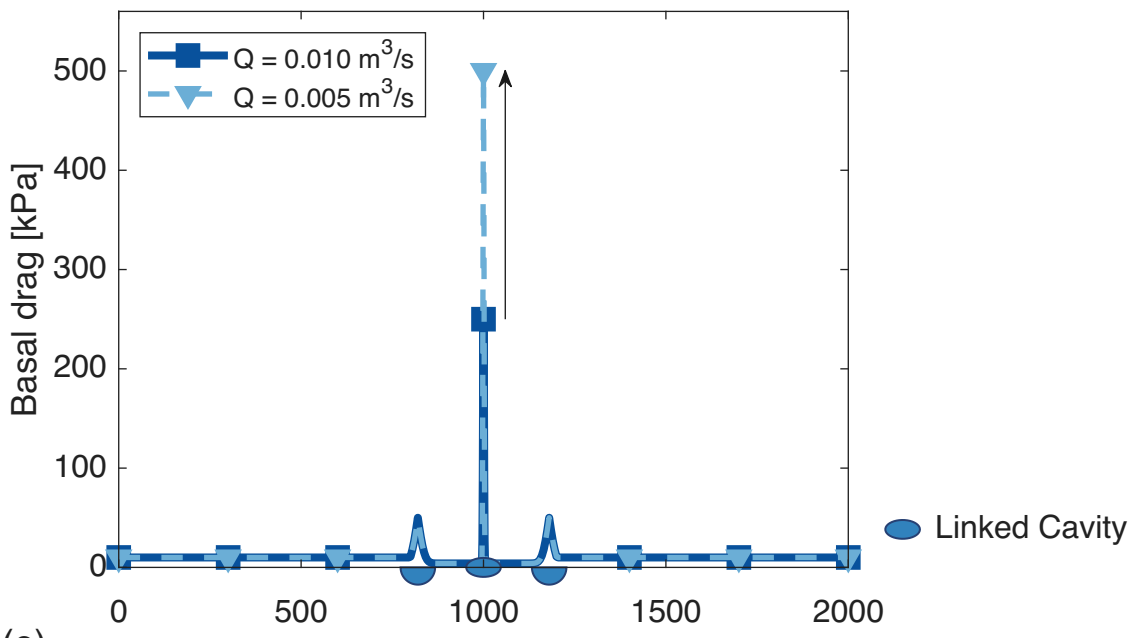
(b)


Q = Q0 m3/s
Q = 0.512 Q0 m3/s
Ice surface speed [m/yr]
0
20
40
60
80
100
120
140
160
0
500
1000
1500
2000
No change

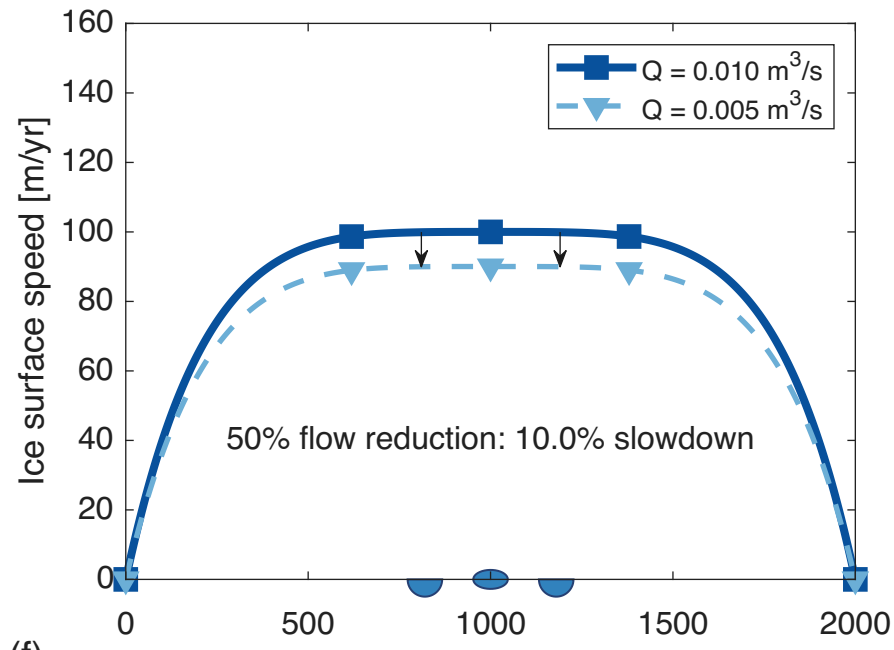
(c)

(d)

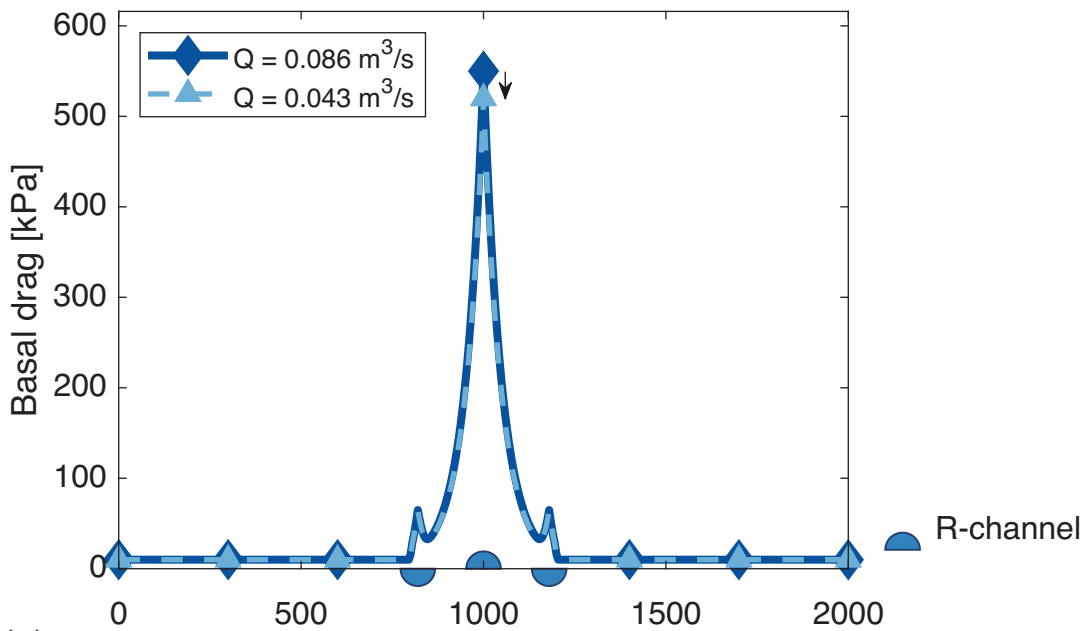
(e)


Q = 0.010 m3/s
Q = 0.005 m3/s
Basal drag [kPa]
0
100
200
300
400
500
0
500
1000
1500
2000
Linked Cavity

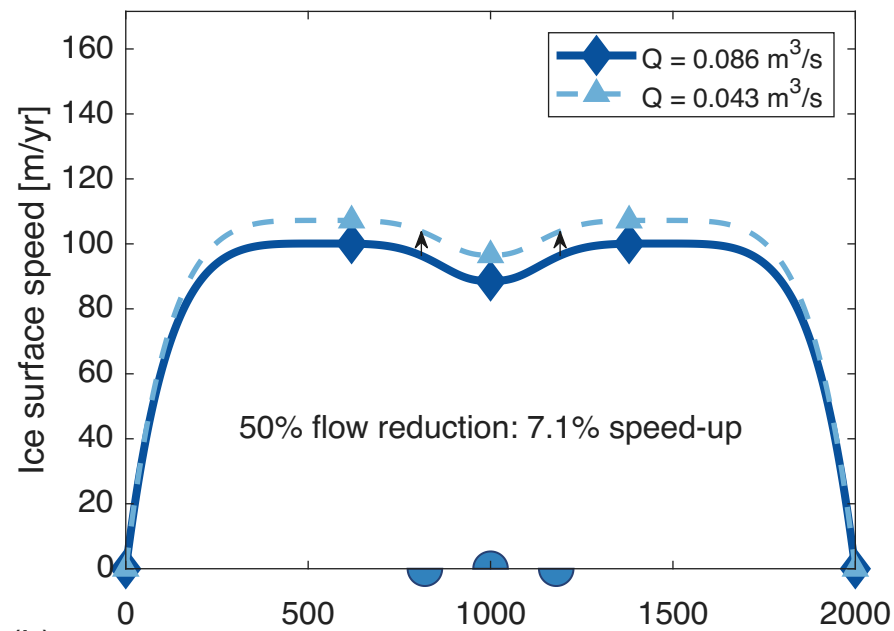
(f)

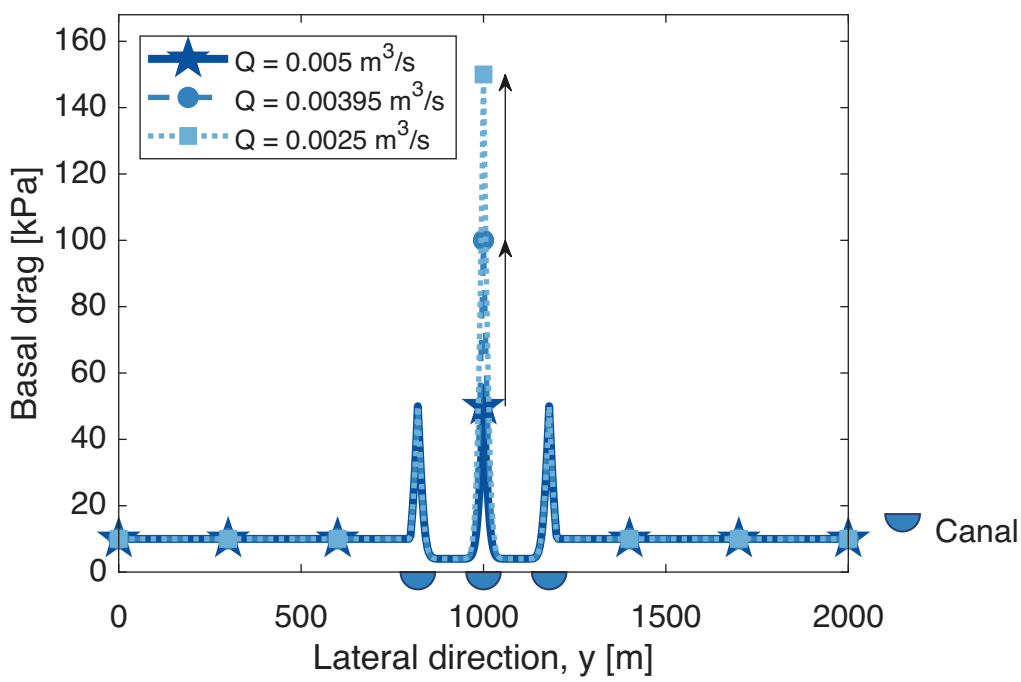

Q = 0.010 m3/s
Q = 0.005 m3/s
Ice surface speed [m/yr]
0
20
40
60
80
100
120
140
160
0
500
1000
1500
2000
50% flow reduction: 10.0% slowdown

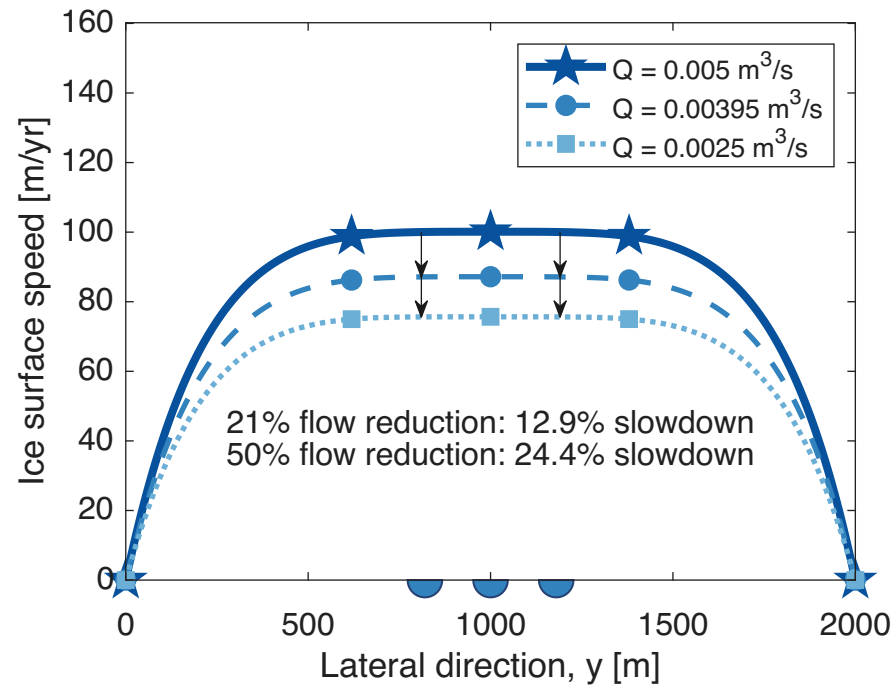
(g)


Q = 0.086 m3/s
Q = 0.043 m3/s
Basal drag [kPa]
0
100
200
300
400
500
600
0
500
1000
1500
2000
R-channel

(h)


Q = 0.086 m3/s
Q = 0.043 m3/s
Ice surface speed [m/yr]
0
20
40
60
80
100
120
140
160
0
500
1000
1500
2000
50% flow reduction: 7.1% speed-up


Q = 0.005 m3/s
Q = 0.00395 m3/s
Q = 0.0025 m3/s
Basal drag [kPa]
0
20
40
60
80
100
120
140
160
0
500
1000
1500
2000
Lateral direction, y [m]
Canal


Q = 0.005 m3/s
Q = 0.00395 m3/s
Q = 0.0025 m3/s
Ice surface speed [m/yr]
0
20
40
60
80
100
120
140
160
0
500
1000
1500
2000
21% flow reduction: 12.9% slowdown
50% flow reduction: 24.4% slowdown
Lateral direction, y [m]